DEVELOPING AN EMOTION DETECTION SYSTEM IN SPEECH FOR ENHANCED REAL-TIME APPLICATIONS

A PROJECT REPORT SUBMITTED IN PARTIAL FULFILLMENT
OF THE REQUIREMENT FOR THE DEGREE OF
BACHELOR OF TECHNOLOGY
IN
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
SUBMITTED BY

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OVERVIEW

- Problem Statement
- What is SER?
- Dataset description
- Algorithm used to recognition
- What have we done yet?
- Benefits
- Use Cases

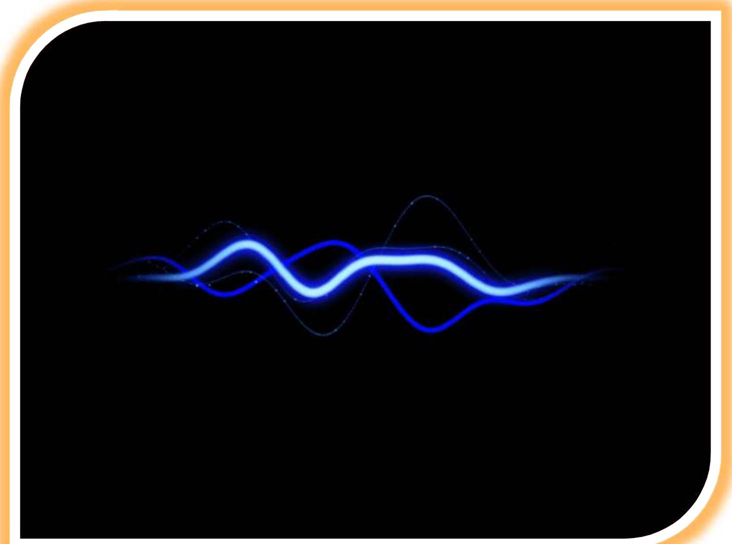
PROBLEM STATEMENT

Develop a real-time emotion detection system from speech, addressing key challenges like data variability, cultural and contextual adaptation, and latency in processing.

WHAT IS SER?

- Instantly captures and analyzes vocal nuances to detect emotions as they happen, ensuring responsive and adaptive interactions.

- **Speech Emotion Recognition (SER)** primarily focuses on analyzing vocal features from audio signals, so it relies more on **signal processing** and **audio-based** machine learning techniques



DATASET DESCRIPTION

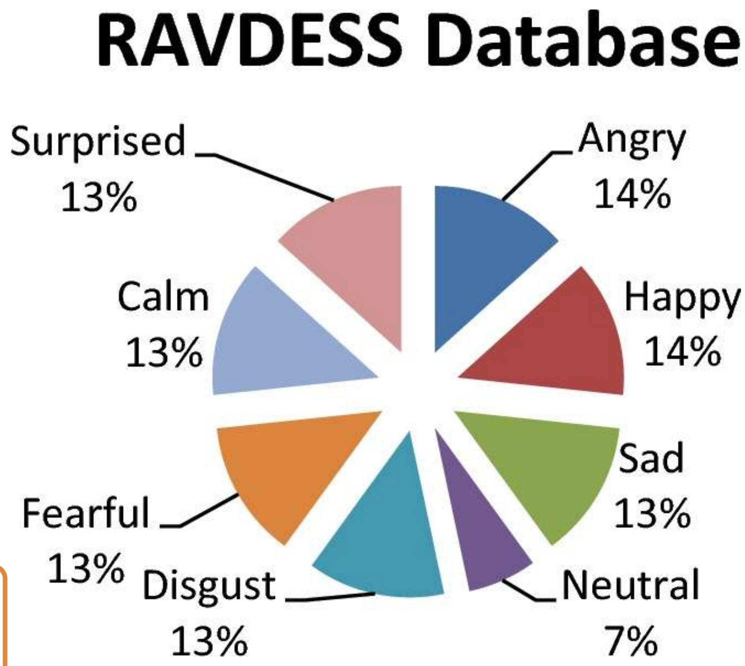
Dataset Used: **RAVDESS (Ryerson Audio-Visual Database of Emotional Speech and Song)**

Content Overview: Actors: 24 professional actors (12 male, 12 female).

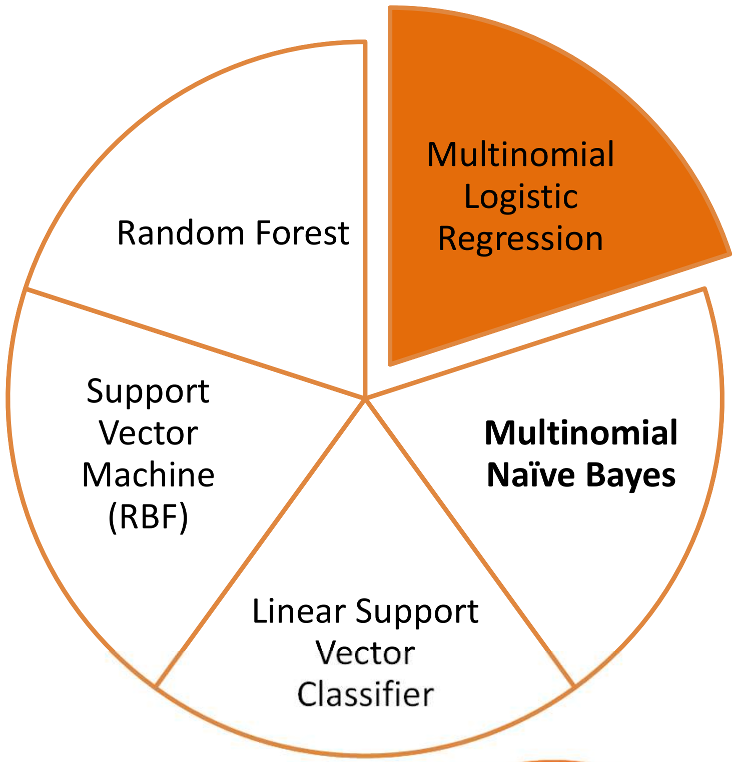
Emotions Covered: Happy, sad, angry, fearful, calm, surprise, disgust and neutral.

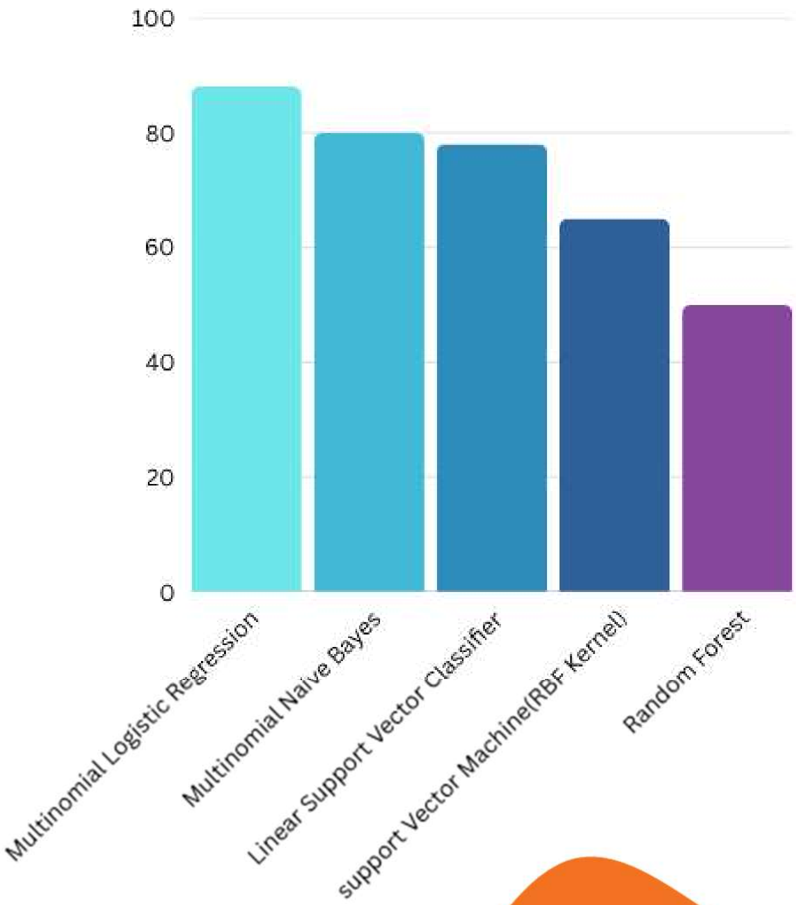
File Format: A RAVDESS filename consists of a 7-part numerical identifier (e.g., 03-01-06-01-02-01-12.wav).

Total Samples: 1,440 speech recordings with various emotional expressions.



MULTI-MODAL COMPARISON CHART

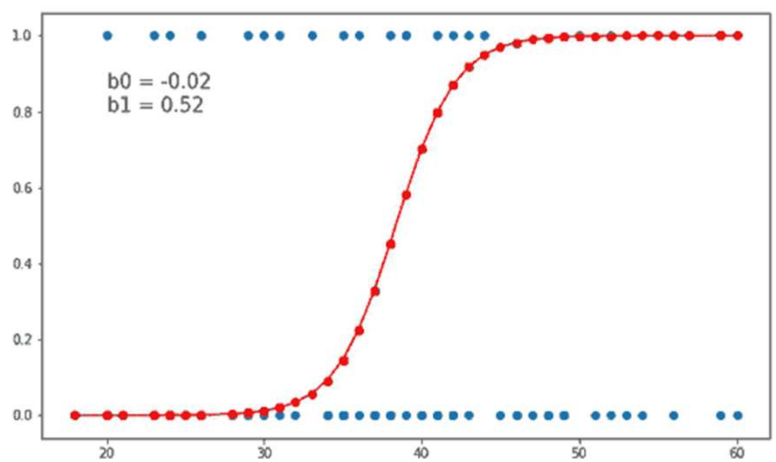


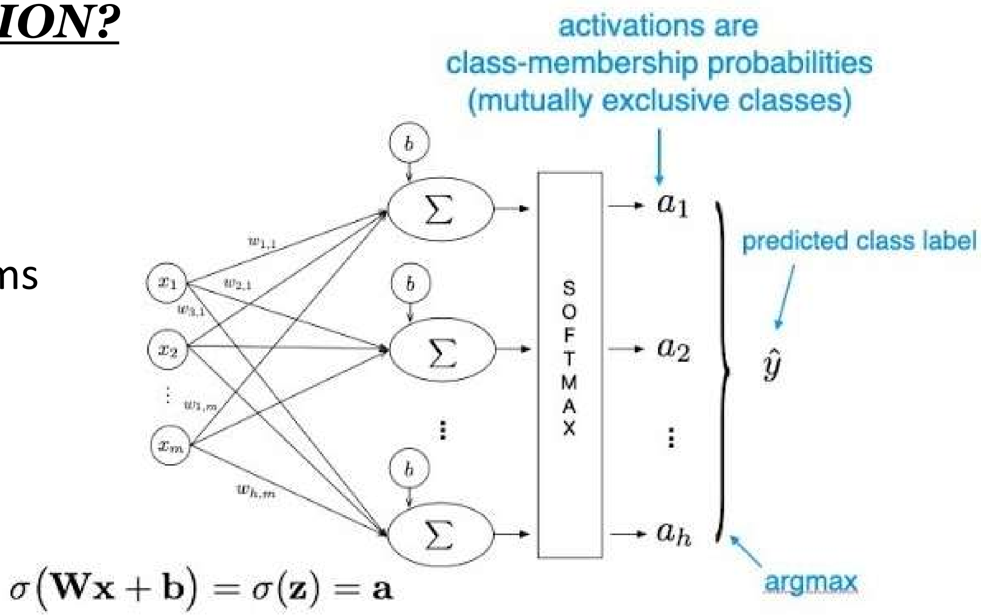


ML ALGORITHM USED FOR SER

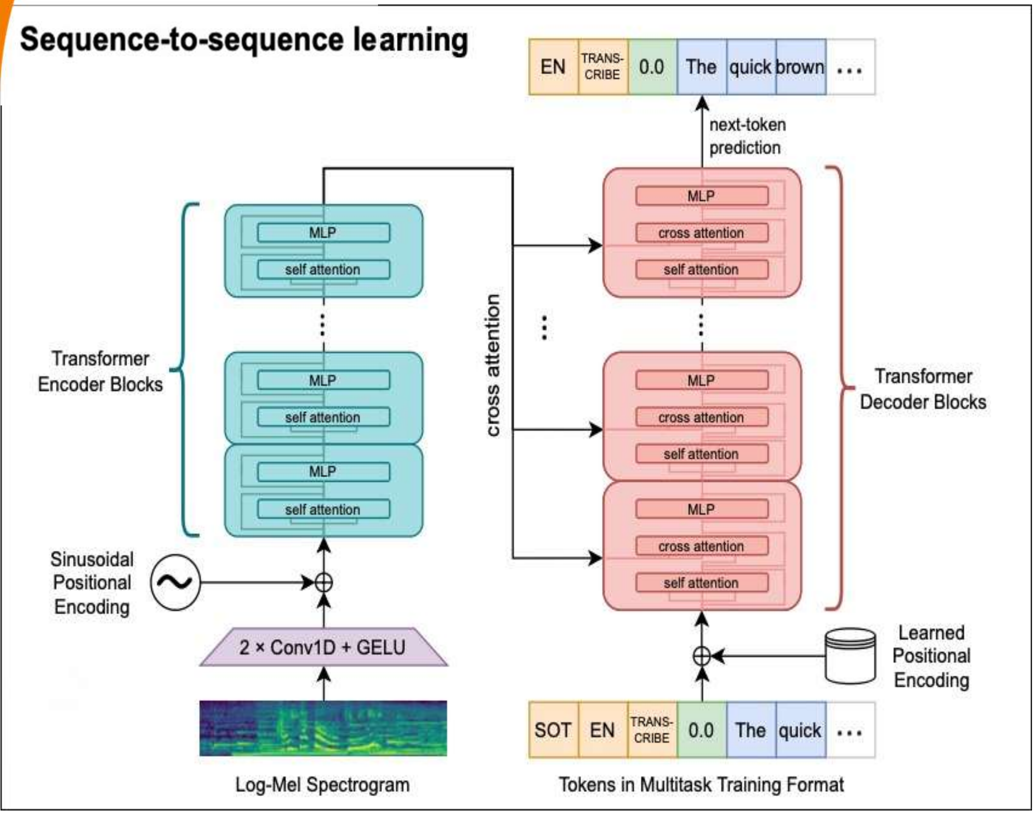
WHY MULTINOMIAL LOGISTIC REGRESSION?

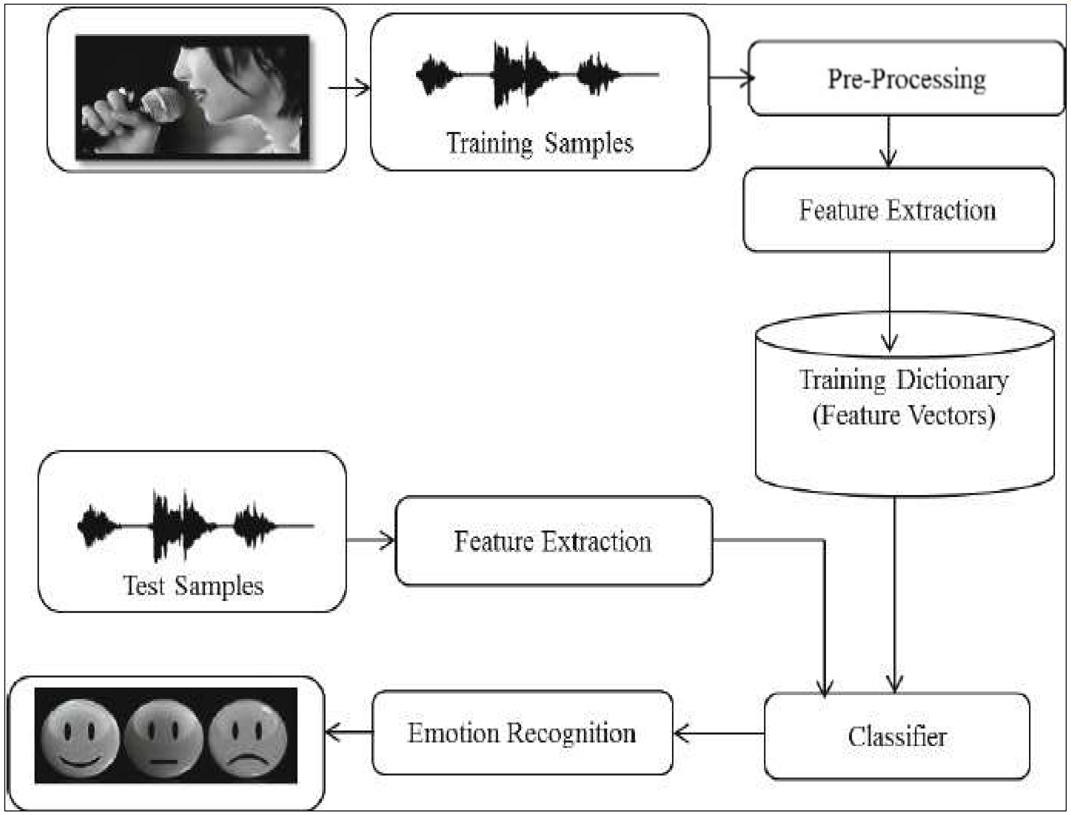
- Easy to implement and understand
- Acts as a clear and interpretable baseline model
- Delivers reliable and efficient performance
- Suitable for solving multi-class classification problems





SER-ML PROJECT PIPELINE





❖ ***Open-AI Whisper-Base Model Architecture***

❖ ***Multinomial LR Pipeline***

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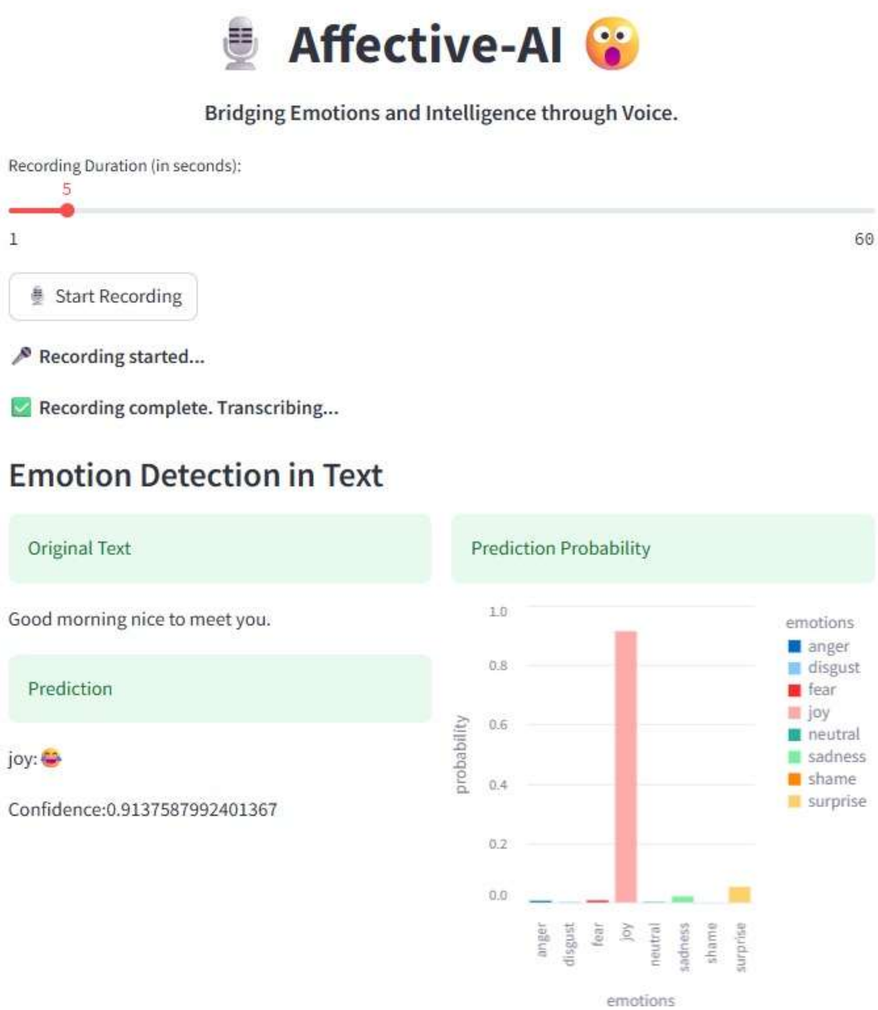
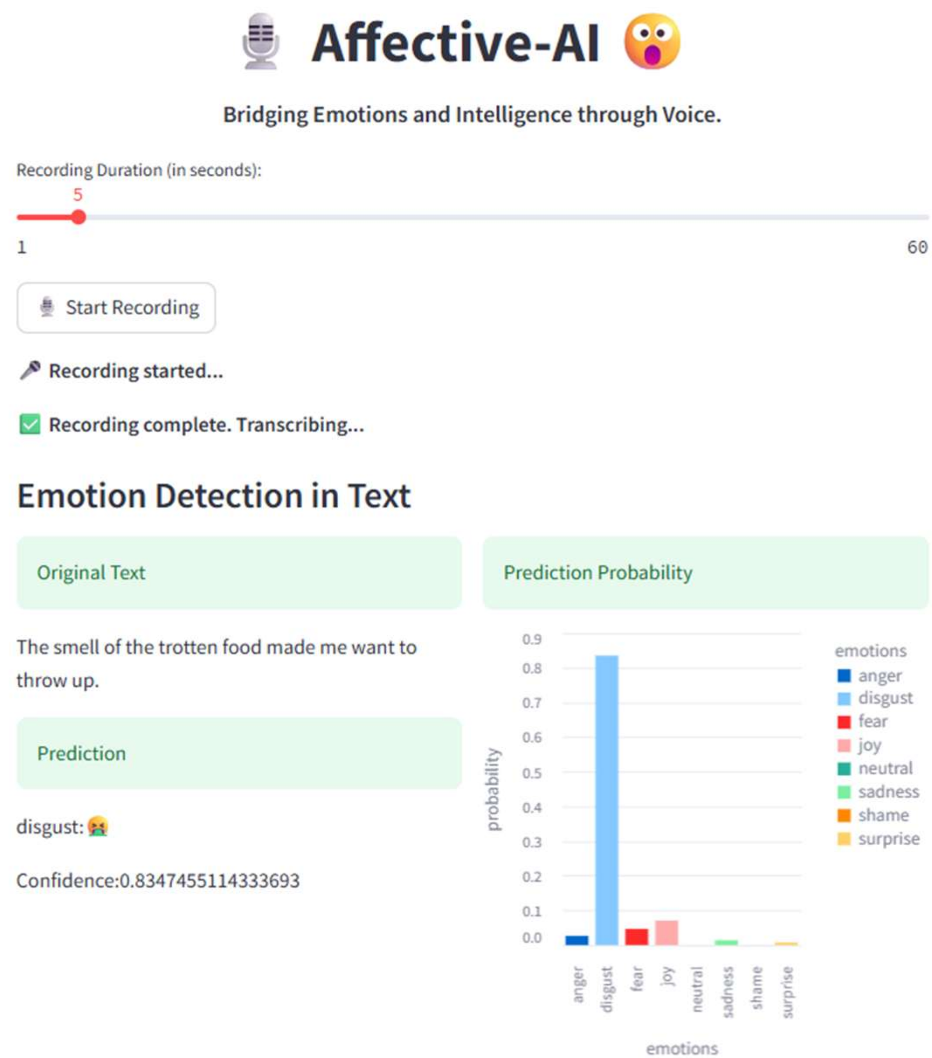

Affective-AI
Bridging Emotions and Intelligence through Voice.
Recording Duration (in seconds):
5
1
60
Start Recording
Recording started...
Recording complete. Transcribing...
Emotion Detection in Text
Original Text
Good morning nice to meet you.
Prediction
joy:
Confidence:0.9137587992401367
Prediction Probability
probability
emotions
anger
disgust
fear
joy
neutral
sadness
shame
surprise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Bridging Emotions and Intelligence through Voice.
Recording Duration (in seconds):
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Start Recording
Recording started...
Recording complete. Transcribing...
Emotion Detection in Text
Original Text
The smell of the trotten food made me want to throw up.
Prediction
disgust:
Confidence:0.8347455114333693
Prediction Probability
probability
emotions
anger
disgust
fear
joy
neutral
sadness
shame
surprise

BENEFITS

- Understand user emotions quickly
- Make better decisions with emotion data
- Fast and real-time analysis
- Gain insights into customer behavior
- Easy to use for everyone

USE CASES AND APPLICATIONS

- Customer Support Monitoring
- Mental Health Companion
- E-learning Feedback
- Voice-Based Product Reviews
- Gaming Experience Enhancement

FUTURE ADVANCEMENTS

- Multilingual & Personalized Features
- Smarter Emotion Insights
- Real-Time & Mobile Optimization
- Cross-Modal Emotion Detection
- SaaS Launch & API Integration

THANK YOU!!!!